Thirunagari Sreenith

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.88
NARAYANA JR COLLEGE	Class XII (TSBIE)	2023	96.40%
NARAYANA SCHOOL	Class X (SSC)	2021	10.00%

PROJECTS

STAAD.PRO - based building design using Reinforced Concrete

- Modelled and analysed a multi-span RCC frame using STAAD.Pro under gravity loads and load combinations for structural design and evaluated behaviour of the frame under different loading conditions and design requirements.
- Designed one-way slabs, three-span beams, and columns considering flexure, shear, and combined axial load-moment effects to ensure structural performance and appropriate reinforcement requirements for the designed RCC members.
- Developed reinforcement detailing and structural design drawings in accordance with IS 456 and SP 34 provisions for RCCmembers, applying standard design requirements to prepare practical and properly detailed structural components.

Workability of MRFA-Based High-Performance Concrete

- Investigated the effects of additional water and superplasticizer on slump flow and slump loss of HPC using 100% MRFA with additional water levels of 0%, 3%, and 6% and superplasticizer dosages of 2%, 2.5%, and 3%.
- Designed and conducted 9 concrete mixes with 3 replications, resulting in 27 experimental runs using a face-centred central composite design, and evaluated slump flow at different time intervals to study workability and retention.
- Developed a second-order response surface model to analyse slump flow behaviour, achieving a high coefficient of determination($R^2 = 0.943$) and identifying additional water as the major factor influencing the workability of HPC.

3D Printed Model of Petronas Towers

- Designed and developed a detailed 3D model of the Petronas Twin Towers using Solid Edge with accurate dimensions and features, maintaining architectural accuracy and appropriate scale for producing a high-fidelity miniature model.
- Prepared the model for 3D printing using Cura by optimizing support structures and minimizing material consumption during the printing process while ensuring proper geometry, printability, and overall quality of the final printed model.
- Gained practical experience in CAD modelling, 3D printing preparation, slicing techniques, and post-processing through the complete development of the miniature structure from digital modelling to the final physical printed model.

Cost-Benefit & Sustainability Analysis: Automatic Water Faucets

- Analysed water-usage data from a controlled two-phase faucet comparison study, quantifying 36% and 28% reductions in consumption in men's and women's restrooms, respectively, after switching from manual to automatic faucets.
- Developed a multi-variable cost-benefit analysis incorporating water, sewer, installation, maintenance, and battery costs, recalculating ROI using current market rates and reducing the estimated payback period from 74 to 15 years.
- Evaluated faucet sustainability across environmental, social, and economic dimensions and mapped findings to 2 UN SDGs: SDG 12 (Responsible Consumption & Production) and SDG 11 (Sustainable Cities & Communities).

SKILLS

- **Tools**: Latex, Solid Edge, QGIS, STAAD, MS EXCEL,AUTOCAD.
- **SOFT SKILLS**: Adaptability, Problem-Solving, Teamwork, Mentoring.

- **Programming Skills**: C/C++.

RELEVANT COURSES

- **Mathematics**: Calculus, Introduction to Probability, Transform Techniques, Complex Variables, Introduction to Statistics, Differential Equations, Elementary Linear Algebra.
- **Core Courses**: Structural Analysis, Reinforced Concrete Design, Structural Steel Design, Mechanics of Solids and Structures, Geotechnical Engineering, Foundation Engineering, Fluid Mechanics, Hydraulic Engineering, Highway Design and Materials, and Environmental Engineering.

- **CSE & AI**: Introduction to Programming, Introduction to Artificial Intelligence.

POSITIONS OF RESPONSIBILITY

Sunshine Mentor

- Mentored mentees to help them adapt to college life and have a smooth first-year experience.

- Guided mentees on academics, campus life, and day-to-day challenges.

- Maintained regular communication with mentees and provided support whenever needed.

- Volunteered in various Sunshine events and student activities.

EXTRACURRICULAR

National Service Scheme (NSS)

- Completed 120 hours of NSS community service activities.

- Actively participated in various college NSS activities promoting social responsibility.

- Contributed to community service initiatives and social awareness activities.